Lecture 19 – Introduction to large sample theory

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So far, we have studied estimators and hypothesis tests for a fixed sample size n. In some models, we can find the exact distribution of a statistic and use it to answer questions about estimation and testing. When this distribution is difficult to obtain, large sample theory gives us another way forward: we study what happens as $n \to \infty$, and use the limiting behavior to approximate what happens for large n.

The large n case is called the *asymptotic* setting. We need to consider *sequences* of random variables, often denoted by

$$Y_1, Y_2, Y_3, \ldots \qquad \text{or} \qquad \{Y_n\}_{n=1}^{\infty} \qquad \text{or simply} \qquad \{Y_n\}.$$

For example, Y_n might represent the sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$ when the sample size is n. We first formalize what it means for random variables to converge.

19.1 Modes of convergence

Definition 1. A sequence of numbers $a_1, a_2, a_3, \ldots$ **converges** to a if the following holds:

$$\forall \epsilon > 0, \quad \exists N_\epsilon < \infty, \text{ such that } \quad \forall n \geq N_\epsilon, \quad |a_n - a| \leq \epsilon$$

Intuitively, the definition says that eventually, a_n is as close to a as you want it to be: if you set any error tolerance, then for large enough n, you are guaranteed to have a_n within that tolerance of a. Typically the smaller your error tolerance, the larger n you will need to reach it.

Convergence is denoted by

$$\lim_{n\to\infty} a_n = a, \qquad \text{or} \qquad a_n \xrightarrow{n\to\infty} a \qquad \text{or simply} \qquad a_n \to a$$

Example 1. Let's prove that $1/n \to 0$. First consider it intuitively. Set your error tolerance to be 0.01. Is $1/n$ eventually within 0.01 of 0? Yes, once $n > 100$. Is $1/n$ eventually within 0.001 of 0? Also yes, once $n > 1000$.

Now let's do it formally: pick any $\epsilon > 0$ and observe that for all $n \geq 1/\epsilon$, we have $|1/n - 0| = 1/n \leq \epsilon$. Hence $1/n$ converges to 0.

Remark 1. On exams or in homework, I will not ask you to formally prove convergence of a sequence of deterministic numbers.

Definition 2 (Convergence in probability)**.** Let $\{Y_n\}$ be a sequence of random variables. We say that Y_n **converges in probability** to a random variable Y if

$$\forall \delta > 0, \quad \lim_{n\to\infty} \mathbb{P}(|Y_n - Y| > \delta) = 0$$

Convergence in probability is denoted by $Y_n \xrightarrow{p} Y$. The definition essentially says that it becomes increasingly rare for Y_n to be some distance δ from Y. This is true even if δ is very small! So for example $\mathbb{P}(|Y_n - Y| > 10^{-3})$ still converges to zero.

Example 2. Suppose $Y_n \sim \mathsf{Bernoulli}\big(p/n\big)$ for $n = 1, 2, 3, \ldots$. Let's prove that $Y_n \xrightarrow{p} 0$. To do this, we need to take any $\delta > 0$ and show $\mathbb{P}(|Y_n - 0| > \delta)$ converges to zero. Let's consider two cases. If $\delta \geq 1$, then $\mathbb{P}(|Y_n| > \delta) = \mathbb{P}(Y_n < -\delta) + \mathbb{P}(Y_n > \delta) = 0$, since Y_n can't be negative and Y_n can't be greater than 1. Therefore $\mathbb{P}(|Y_n| > \delta)$ converges to zero. Next suppose $0 < \delta < 1$. Then

$$\mathbb{P}(|Y_n - 0| > \delta) = \mathbb{P}(Y_n > \delta) = \mathbb{P}(Y_n = 1) = \frac{p}{n} \to 0.$$

So in both cases, we have shown the necessary convergence to zero.

Definition 3 (Convergence in distribution)**.** Let $\{Y_n\}$ be a sequence of random variables, each with distribution function F_n, and let Y be a random variable with distribution function F. We say Y_n **converges in distribution** to Y if

$$\lim_{n\to\infty} F_n(y) = F(y)$$

for every y such that F is continuous at y. Convergence in distribution is denoted by $Y_n \xrightarrow{d} Y$.

Example 3. Let $Y_n \sim \mathsf{Unif}(0, 1/n)$. Let's show that $Y_n \xrightarrow{d} 0$. The cdf of $Y = 0$ is a step function: $F(t) = 0$ for all $t < 0$ and $F(t) = 1$ for all $t \geq 0$. First consider any $t < 0$. Then $F_n(t) = 0$ as well, and 0 converges to zero. Next consider $t > 0$. Take n large enough that $1/n < t$. For all such n, we have $F_n(t) = 1$, and 1 converges to 1. Thus we have checked that $F_n(t)$ converges to $F(t)$ for all t where $F(t)$ is continuous, so $Y_n \xrightarrow{d} 0$. Let's see what happens at $t = 0$. For all n, we have $F_n(0) = 0$, but $F(0) = \mathbb{P}(Y = 0 \leq 0) = 1$. So $F_n(0)$ does not converge to $F(0)$, but this does not contradict $Y_n \xrightarrow{d} 0$, since we don't have to check convergence of cdf's at $t = 0$, since F is discontinuous there.

Example 4. Suppose $X_i \overset{\text{iid}}{\sim} \mathsf{Unif}(0, 1)$, and let $Y_n = X_{(n)} = \max\{X_1, \ldots, X_n\}$. Show that $Y_n \xrightarrow{d} 1$. For $0 \leq t \leq 1$, the distribution function of Y_n is

$$F_n(t) = \mathbb{P}(Y_n \leq t) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq t) = t^n$$

Thus the sequence of distribution functions converges to a step function at 1,

$$\lim_{n\to\infty} F_n(t) = \begin{cases} 0, & t < 1, \\ 1, & t \geq 1. \end{cases}$$

This is exactly the distribution function of the constant "random variable" equal to 1. Hence

$$Y_n = X_{(n)} \xrightarrow{d} 1.$$

Theorem 1. Convergence in probability implies convergence in distribution, i.e.,

$$Y_n \xrightarrow{p} Y \quad \Longrightarrow \quad Y_n \xrightarrow{d} Y$$

but the reverse implication does not hold in general. In the special case where $Y = c$ is a constant, the reverse implication *does* hold, so we conclude that convergence in probability is *equivalent* to convergence in distribution.

Proof that $Y_n \xrightarrow{d} c$ implies $Y_n \xrightarrow{p} c$. Let F_n be the cdf of Y_n, and fix any $\delta > 0$. We need to show $\mathbb{P}(|Y_n - c| > \delta)$ converges to zero. But

$$\begin{aligned} \mathbb{P}(|Y_n - c| > \delta) &\leq \mathbb{P}(Y_n \leq c - \delta) + \mathbb{P}(Y_n > c + \delta) \\ &= F_n(c - \delta) + 1 - F_n(c + \delta) \to 0 + 1 - 1 = 0, \end{aligned} \tag{19.1}$$

since $c - \delta$ and $c + \delta$ are continuity points of the cdf of the constant c. □

Example 5 (Convergence need not preserve expectations)**.** $Y_n \xrightarrow{p} Y$ does not guarantee that $\mathbb{E}[Y_n] \to \mathbb{E}[Y]$ (and therefore $Y_n \xrightarrow{d} Y$, which is weaker than convergence in probability, also does not guarantee convergence of means). To see why consider the following example:

$$Y_n = \begin{cases} Y, & \text{if } X_n = 1, \\ n^2, & \text{if } X_n = 0, \end{cases} \tag{19.2}$$

where $X_n \sim \text{Bernoulli}(1 - 1/n)$, and X_n is independent of Y. Assume $\mathbb{E}[|Y|] < \infty$.

To prove convergence in probability, fix any $\delta > 0$. Since $Y_n = Y$ whenever $X_n = 1$, the event $\{|Y_n - Y| > \delta\}$ can occur only when $X_n = 0$. Hence

$$\mathbb{P}(|Y_n - Y| > \delta) \leq \mathbb{P}(X_n = 0) = \frac{1}{n} \to 0.$$

Thus $Y_n \xrightarrow{p} Y$.

On the other hand, using independence of X_n and Y,

$$\mathbb{E}[Y_n] = \left(1 - \frac{1}{n}\right)\mathbb{E}[Y] + \frac{1}{n}\, n^2 = \left(1 - \frac{1}{n}\right)\mathbb{E}[Y] + n \to +\infty.$$

Therefore $\mathbb{E}[Y_n]$ does not converge to the finite value $\mathbb{E}[Y]$. The exceptional value n^2 occurs with probability tending to zero, but its contribution to the expectation is n, which grows without bound.

19.2 Operations which preserve convergence

Theorem 2 (Continuous Mapping Theorem)**.** Let $\{Y_n\}$ be a sequence of random variables such that $Y_n \xrightarrow{p} Y$, and let g be a continuous function. Then

$$g(Y_n) \xrightarrow{p} g(Y).$$

Theorem 3 (Convergence of sums and products)**.** If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$ then $X_n + Y_n \xrightarrow{p} X + Y$ and $X_n Y_n \xrightarrow{p} XY$.

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ then $X_n + Y_n \xrightarrow{d} X + c$ and $X_n Y_n \xrightarrow{d} Xc$.

The second statement is known as Slutsky's Theorem.

Remark 2. In general, $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ does *not* imply $X_n + Y_n \xrightarrow{d} X + Y$ or $X_n Y_n \xrightarrow{d} XY$. In fact, a statement like this does not even make sense, without additional information about the joint distribution of X_n and Y_n, and the joint distribution of X and Y.

For example, take a sequence of X_n's such that $X_n \sim \mathcal{N}(0, 1)$, and let $Y_n = -X_n$. By symmetry of the standard normal, $Y_n \sim \mathcal{N}(0, 1)$ as well. Therefore, we can take any two random variables X, Y which both have the $\mathcal{N}(0, 1)$ distribution, and conclude that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$.

So does $X_n + Y_n = 0$ converge in distribution to $X + Y$? It depends on the relationship between X and Y! If we take $Y = -X$, then $X + Y = 0$, and 0 indeed converges to 0. If we take $Y = X$, then $X + Y = 2X \sim \mathcal{N}(0, 4)$, and 0 does not converge to $\mathcal{N}(0, 4)$.

It is common to write the limit of a convergence in distribution using the law itself, rather than a random variable. So in this example, we would write

$$X_n \xrightarrow{d} \mathcal{N}(0, 1), \qquad Y_n \xrightarrow{d} \mathcal{N}(0, 1).$$

This makes it clear that we don't have enough information to know what happens when we add the two $\mathcal{N}(0, 1)$'s.

19.3 Limit theorems

Theorem 4 (Law of Large Numbers)**.** Let $\{X_n\}$ be a sequence of independent and identically distributed (i.i.d.) random variables with finite expectation $\mathbb{E}[X_1] = \mu$. Then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu.$$

Example 6. Suppose $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

Then the sample mean $\bar{X}_n$ is an estimator for μ. Does it converge to the true value?

Since $\mathbb{E}[X_i] = \mu$, the LLN implies

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu.$$

An estimator that converges in probability to the true parameter is called *consistent*. Thus, $\bar{X}_n$ is a consistent estimator for μ.

Example 7. Suppose $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$

From the LLN we know that the sample mean converges in probability to the population mean:

$$\bar{X}_n \xrightarrow{p} \mu$$

For the continuous function $g(x) = x^2$ it follows that

$$g(\bar{X}_n) = \bar{X}_n^2 \xrightarrow{p} \mu^2$$

This means that $(\bar{X}_n)^2$ is a *consistent estimator* of μ^2.

Remark 3. An estimator $\hat{\theta}_n$ of θ based on n samples, whose bias converges to zero as $n \to \infty$, is called **asymptotically unbiased**. This is the statement that $\text{bias}_\theta(\hat{\theta}_n) = \mathbb{E}_\theta[\hat{\theta}_n] - \theta$ converges to zero. Equivalently, $\mathbb{E}_\theta[\hat{\theta}_n] \to \theta$, $n \to \infty$. As discussed above, convergence in probability does not imply convergence of expectations. Therefore, consistency of an estimator does not guarantee asymptotic unbiasedness of the estimator.

Theorem 5 (Central Limit Theorem)**.** Let $\{X_n\}$ be a sequence of iid random variables with $\mathbb{E}[X_1] = \mu$ and $\mathsf{Var}(X_1) = \sigma^2$. Then

$$\sqrt{\frac{n}{\sigma^2}}\left(\bar{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}(0, 1).$$

In other words,

$$\lim_{n\to\infty} \mathbb{P}\left(\sqrt{\frac{n}{\sigma^2}}\left(\bar{X}_n - \mu\right) \le t\right) = \Phi(t) \coloneqq \mathbb{P}(Z \le t), \qquad Z \sim \mathcal{N}(0, 1)$$

Remark 4. For the purposes of statistical inference, an important implication of the CLT is that we can approximate the distribution of the sample mean (for general statistical models) with a normal distribution.

Example 8. Suppose $X_i \overset{\text{iid}}{\sim} f_\theta(x)$ where the statistical model $\{f_\theta(x) \mid \theta \in \Theta\}$ is unknown or difficult to work with. Let $\mu = \mathbb{E}_\theta[X_1]$ and $\sigma^2 = \mathsf{Var}_\theta[X_1]$ be the mean and variance of X_1. Consider the testing problem,

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu > \mu_0$$

For the normal model $\mathcal{N}(\mu, \sigma^2)$ we know that the uniformly most powerful test of level α is given by

$$\mathcal{R} = \{\underline{x} \ : \ \bar{x}_n > \kappa\},$$

where κ is chosen so that $\mathbb{P}_{\mu_0}[\bar{X}_n > \kappa] = \alpha$. In the normal model, we know $\sqrt{\frac{n}{\sigma^2}}(\bar{X}_n - \mu_0) \sim \mathcal{N}(0,1)$, so we can directly determine $\kappa = \mu_0 + \frac{\sigma}{\sqrt{n}} z_{1-\alpha}$.

If we want to use this test for our setting, how should we choose the threshold κ?

One option is to appeal to the CLT. As $n \to \infty$, we have

$$\begin{aligned}\mathbb{P}_{\mu_0}[\bar{X}_n > \kappa] &= \mathbb{P}_{\mu_0}\Big[\sqrt{\tfrac{n}{\sigma^2}}(\bar{X}_n - \mu_0) > \sqrt{\tfrac{n}{\sigma^2}}(\kappa - \mu_0)\Big] \\ &\to \mathbb{P}_{\mu_0}\Big[Z > \sqrt{\tfrac{n}{\sigma^2}}(\kappa - \mu_0)\Big]\end{aligned}$$

where $Z \sim \mathcal{N}(0,1)$.

Specifying κ according to the normal approximation yields

$$\begin{aligned}\mathbb{P}\Big[Z > \sqrt{\tfrac{n}{\sigma^2}}(\kappa - \mu_0)\Big] = \alpha \iff \sqrt{\tfrac{n}{\sigma^2}}(\kappa - \mu_0) &= z_{1-\alpha} \\ \iff \kappa &= \mu_0 + \frac{\sigma}{\sqrt{n}} z_{1-\alpha}.\end{aligned}$$

This threshold results in a test with *asymptotic level* α.

Remark 5. One potential problem with the test in the previous example is that the threshold depends on variance σ^2, which is typically unknown. A common approach used in practice is to replace σ^2 with an estimate provided by the sample variance

$$S_n^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$$

The reasonableness of this approach can be justified using Slutsky's theorem.

Example 9. Suppose $\{X_i\}$ are iid with $\mathbb{E}[X_1] = \mu$ and $\mathsf{Var}(X_1) = \sigma^2$ both unknown.

By the CLT,

$$\sqrt{n}\big(\bar{X}_n - \mu\big) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

By the LLN and continuous mapping theorem,

$$\begin{aligned}S_n^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2 &= \frac{1}{n}\sum_{i=1}^n X_i^2 - \bar{X}_n^2 \\ &\xrightarrow{p} \mathbb{E}[X_1^2] - \mu^2 = \sigma^2.\end{aligned}$$

Applying Slutsky's Theorem, we conclude that

$$\sqrt{\frac{n}{S_n^2}}(\bar{X}_n - \mu) \xrightarrow{d} Z \sim \mathcal{N}(0,1).$$

Example 10. We return to the testing problem where $X_i \overset{\text{iid}}{\sim} f_\theta(x)$ with unknown mean $\mathbb{E}[X_i] = \mu$ and unknown variance $\mathsf{Var}(X_i) = \sigma^2$, and the hypotheses are

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

Previously, we considered a test with asymptotic level α based on thresholding the sample mean

$$\mathcal{R} = \left\{ \underline{x} : \bar{x}_n > \mu_0 + \sqrt{\frac{\sigma^2}{n}} z_{1-\alpha} \right\}$$

But this threshold depends on σ^2, which is unknown!

One solution is to use the sample variance S_n^2 as a replacement for σ^2. As $n \to \infty$, we have

$$\begin{aligned} \mathbb{P}_{\mu_0}\left[\bar{X}_n > \mu_0 + \sqrt{\frac{S_n^2}{n}} z_{1-\alpha} \right] &= \mathbb{P}_{\mu_0}\left[\sqrt{\frac{n}{S_n^2}} (\bar{X}_n - \mu) \geq z_{1-\alpha} \right] \\ &\to \mathbb{P}(Z \geq z_{1-\alpha}) = \alpha \end{aligned}$$

where $Z \sim \mathcal{N}(0,1)$.

Thus we have constructed a test with asymptotic level α that does not require knowledge of σ^2:

$$\begin{aligned} \mathcal{R} &= \left\{ \underline{x} : \bar{x}_n > \mu_0 + \sqrt{\frac{s_n^2}{n}} z_{1-\alpha} \right\} \\ &= \left\{ \underline{x} : \sqrt{\frac{n}{s_n^2}} (\bar{x}_n - \mu_0) > z_{1-\alpha} \right\}. \end{aligned}$$